HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY111 Course Title: Physics I

Mid-Term Examination Semester: Fall 2023
Date: 12/9/2023 Total Time: 1HR: 30 Mins. **Total Marks: 30**

Answer any three (03) out of the following questions: **$3 \times 10 = 30$**

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

Q1 a. What is meant by wave motion? CLO1 6

Q1 b. A nurse measured the average heart beats of a patient and reported to the doctor in terms of time period as 0.8 sec. Express the heart beat of the patient in terms of number of beats measured per minute. CLO2 4

Q2 a. Write down the properties of longitudinal wave. CLO1 5

Q2 b. Consider a medical imaging device that produces ultrasound by oscillating with a frequency of 2.5×10^6 Hz. What is the time period of this oscillation? CLO2 5

T

Q3. Show that the equation of a plane progressive wave motion in the positive direction of x can be expressed as: $y = a\sin\frac{2\pi}{\lambda}(vt - x)$. CLO2 10

Q4 a. Prove that the differential equation for one dimensional wave can be expressed as $\frac{d^2y}{dt^2} = v^2\frac{d^2y}{dx^2}$. CLO1 6

Q4 b. What is the wavelength of an earthquake that shakes you with a frequency of 10.0 Hz and gets to another city 84.0 km away in 12.0 sec? CLO2 4

5 a. What are phase (or wave) and group velocities? CLO1 4

5 b. Show that the relation between phase and group velocity is: $v_g = v_p - \lambda\frac{dv_p}{d\lambda}$. CLO1 6

(b) For the circuit shown in Fig. 2(b), calculate the voltage v, the conductance G, and the power P. (6) CLO 2

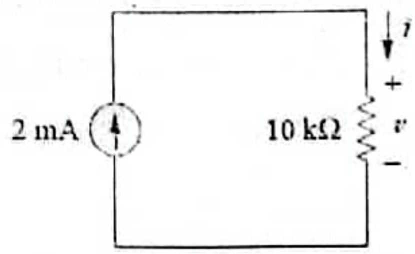

Fig. 2(b)

3. (a) Define node, branch and loop of an electrical circuit. Determine the number of branches and nodes in the circuit shown in Fig. 3(a). Identify the elements that are in series and in parallel. (5) CLO 1

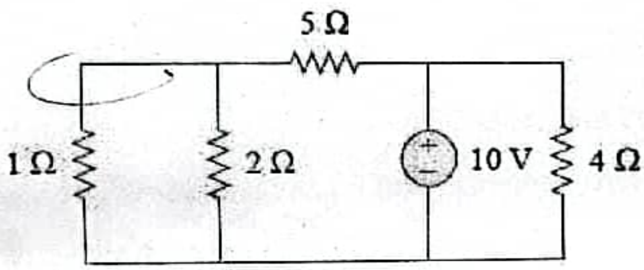

Fig. 3(a)

(b) State Kirchhoff's Voltage and Current Law. Find v_0 and i_0 in the circuit of Fig. 3(b). (5) CLO 2

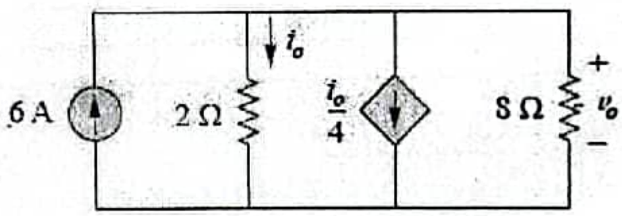

Fig. 3(b)

4. (a) Transform the wye network in Fig. 4(a) to an equivalent delta network. (5) CLO 1

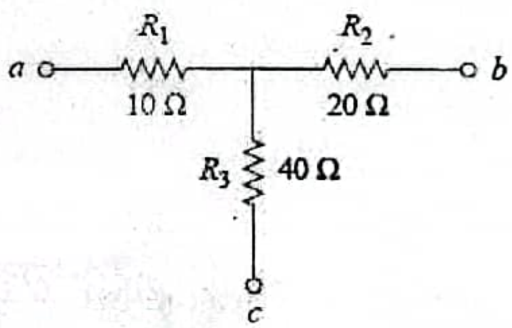

Fig. 4(a)

(b) For the circuit shown in Fig. 4(b), find the node voltages using nodal analysis. (5) CLO 2
Also Find v *and* i in the circuit

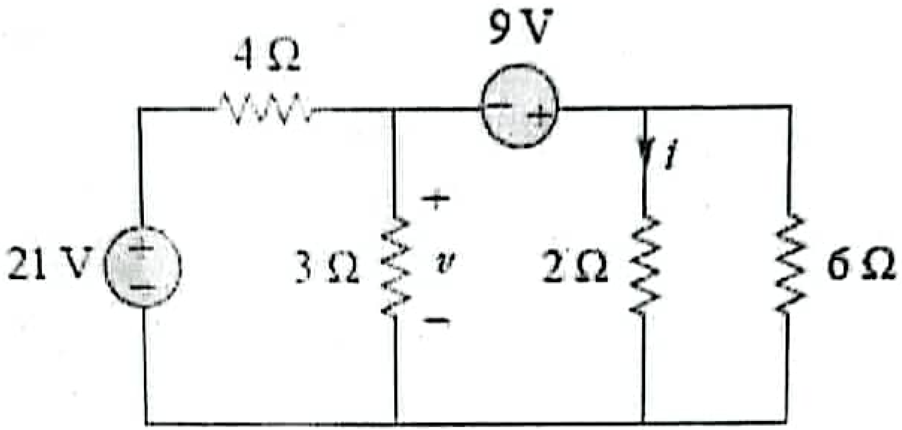

Fig. 4(b)

Hamdard University Bangladesh

Dept. of Electrical and Electronic Engineering (EEE)

Mid-Term Examination (Fall'23)

EEE 117 Electrical Engineering

Time: 1 hour and 30 minutes — Marks: 30

Answer to the question no. 1 (must) and any 2 (two) of the questions among question no. 2, 3 and 4 from the following sets of questions

1. (a) Define the following terms. (4) CLO 1

   i. Electrical circuit
   ii. Electrical current
   iii. Power
   iv. Dependent and Independent source

   (b) The daily electricity consumption chart for a homeowner is given below. (6) CLO 1

Device	Operation time (h)	Rated power (W)	Number of the device
Refrigerator	8	200	1
TV	6	50	1
Computer	4	50	1
Iron	0.25	1600	1
Oven	0.5	1000	1
Lamp	6	25	6
Dishwasher	0.5	1000	1
Washing machine	0.25	1000	1
Air conditioning	6	1000	1

Table 01: Daily Electricity Consumption Chart

Determine the electricity bill in the month of May for the homeowner using the following residential rate schedule:

Base monthly charge of Tk.60.00.

First 100 kWh per month at Tk.5.50 per kWh

Next 100 kWh per month at Tk.5.00 per kWh

Over 200 kWh per month at Tk.4.80 per kWh

2. (a) State Ohm's law. An electric iron draws 2A current at 120V. Find its resistance and the conductance. (4) CLO 1

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering (CSE)
Faculty of Science, Engineering and Technology
1st Year 1st Semester Examination 2023

Course Code: MAT115 | **Course Title: Mathematics I (Differential and Integral Calculus)**
Midterm Examination | Semester: Fall 2023
Date: 12/12/2023 | Total Time: 1HR: 30 Mins. | **Total Marks: 30**

Answer any three (03) out of the following questions: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

			CLOs	Marks
Q1.	**(a)**	Define function, domain and range of a function.	CLO1	[3]
	(b)	Find the domain and range of the following functions:	CLO2	[3]

$$f(x) = \begin{cases} 3 + 2x & \text{when } -\frac{3}{2} \le x < 0 \\ 3 - 2x & \text{when } 0 \le x \le \frac{3}{2} \\ 3 + 2x & \text{when } \frac{3}{2} < x \end{cases}$$

	(c)	Define limit. A function is defined as	CLO2	[4]

$$f(x) = \begin{cases} 1 + 2x & \text{when } -\frac{1}{2} \le x < 0 \\ 1 - 2x & \text{when } 0 \le x \le \frac{1}{2} \\ -1 + 2x & \text{when } \frac{1}{2} < x \end{cases}$$

Does the limit exists at $x = 0$ and $x = \frac{1}{2}$.

Q2.	**(a)**	Find $\frac{dy}{dx}$ of the following functions (any two):	CLO2	[5]

$(i)\ x^y = y^x$

$(ii)\ y = x^x + (\sin x)^{\ln x}$

$(iii)\ y = x^{\sin^{-1} x}$

	(b)	Differentiate (any two):	CLO2	[5]

(i) $x^{\sin^{-1} x}$ with respect to $\sin^{-1} x$

(ii) $\tan^{-1}\left(\frac{2x}{1 - x^2}\right)$ with respect to $\sin^{-1}\left(\frac{2x}{1 + x^2}\right)$

(iii) $\sin^{-1}\left(\frac{2x}{1 + x^2}\right)$ with respect to $\cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$

Q3.	**(a)**	Define continuity and differentiability of a function at any point.	CLO1	[2]
	(b)	A function is defined as follows $f(x) = \begin{cases} 3 + 2x, & \text{when } -3/2 \le x < 0 \\ 3 - 2x, & \text{when } 0 \le x \le 3/2 \\ 3 + 2x, & \text{when } 3/2 \le x \end{cases}$	CLO2	[5]

Discuss the continuity at $x = 0$ and differentiability at $x = 3/2$

	(c)	Show that the minimum value of $\frac{x}{\log x}$ is e.	CLO2	[3]

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: ENG111		Course Title: English Basics
Midterm Examination		Semester: Fall 2023
Date: 12/15/2023	Total Time: 1 hour 30 Mins.	**Total Marks: 30**

Read the following passage and complete the tasks (1—4)

The city of Detroit, in the USA, was once compared to Paris. It had a broad river, smart streets and historically important architecture. Then, in the 20th century, it became 'Motor City'. For a time, most of the world's cars were made here. There was regular work and a good salary in the motor industry. A worker at one of the car factories could own a home, plus a boat, maybe even a holiday cottage. Some say America's middle class was born in Detroit – new highways certainly made it easy for workers to move from the city center to the suburbs in the 1950s. But in the early years of the 21st century, Detroit became America's poorest big city.

In less than five decades the once lively Motor City lost more than half its population. It became known as a city that was failing, full of ruined buildings, extensive poverty and crime. Newspapers and magazines told stories of derelict homes and empty streets. Photographers went to Detroit to record the strange beauty of buildings and city blocks where nature was taking over again. What went wrong in Detroit?

The city is now 69th among US cities for the number of people per square mile. The population fell for several reasons. Partly, it was because people moved to the suburbs in the 1950s. Then there were the shocking riots in 1967, which scared more people away from the city. Then there was the dramatic fall in car manufacture as companies like General Motors and Chrysler faced huge difficulties. And finally, in 2008, came the global financial crisis. Many of Detroit's people are poor – half of the city's families live on less than 25,000 dollars a year.

In 2013, the city did something unusual: it declared itself bankrupt. It was the largest city bankruptcy in US history, at approximately 18-20 billion dollars. Now that the city is free of debt, it has money to do some of what needs to be done. It has replaced about 40,000 streetlights so that places feel safer. The police arrive in answer to calls in less than 20 minutes now, instead of the hour it used to take. And about a hundred empty houses are demolished each week to make space for new buildings. With the nation's biggest city bankruptcy behind it, Detroit is also attracting investors and young adventurers. The New Economy Initiative gave grants of 10,000 dollars to each of 30 new small businesses. It seems that every week a new business opens in Detroit – grocery stores, juice bars, coffee shops, even bicycle makers. Finally, the city is working again.

No.	Question/ Statement	CLOs	Marks
1.	**Answer the following in full sentences:** a) What was the financial state of a random employee at 'Motor City'? b) How many years did it take for Detroit to lose half of its population? c) What does the expression 'nature was taking over again' mean in the text? d) Why did the streets feel unsafe in Detroit formerly? e) What is the frequency of business openings at Detroit now?	CLO2	1x5=5
2.	**Detect whether the statements are true or false. If false, provide the true one:** a) The city of Detroit is in North America. b) Fresh highways were constructed at Detroit from 1946 to 1950. c) The text mentions five reasons for the poverty of Detroit. d) The streets used to feel unsafe owing to darkness. T e) The New Economy Initiative gave grants of total 3 million dollars. F	CLO2	1x5=5

10

3. Rewrite the following passage with right forms of verbs: CLO1 0.5x10=5

made had lived had moved

Before the highways (make), people (live) in the city center. After they (move) to the suburbs, their living expenses (reduce) significantly. However, the city (be) was adversely affected due to this movement. The riots of 1967 (worsen) the situation. The city recently (declare) itself bankrupt. Due to later developments, the city (be) a lot safer now with police (be) able to answer calls more rapidly than before. The city (recover) fully by 2025.

4. Rewrite the following passage by correcting the prepositions where needed: CLO1 0.5x10=5

of

The 1967 Detroit riot was one **in** the bloodiest riots **by** the history **from** the United States. It composed mainly **for** confrontations **between** black residents and the Detroit Police Department. It began **of** Sunday July 23 **at** 1967, **in** Detroit at Michigan. Governor George W. Romney ordered the Michigan Army National Guard **in** help end the disturbance.

5. a) Generate a topic for a persuasive paragraph related with "Urban Life" and then write a 200-words paragraph on it. CLO3 5+5=10

b) Rearrange the following paragraph and make it coherent. Also provide a title for the paragraph.

It is also one of the greatest science fiction and action films ever made. Unaware of his former life, RoboCop executes a campaign against crime. It was directed by Paul Verhoeven and written by Edward Neumeier and Michael Miner. The film has been lauded for its depiction of a robot affected by the loss of humanity, in contrast to the stoic and emotionless robotic characters of that era. Murphy is subsequently revived by the megacorporation Omni Consumer Products as the cyborg law enforcer RoboCop. *RoboCop* is a 1987 American science fiction action film. In the meantime, he also tries to come to terms with the lingering fragments of his humanity. Since its release, *RoboCop* has been hailed as one of the best films of the 1980s. Set in a crime-ridden Detroit in the near future, *RoboCop* centers on police officer Alex Murphy (Weller) who is murdered by a gang of criminals. The film stars Peter Weller, Nancy Allen, Daniel O'Herlihy, Ronny Cox, Kurtwood Smith, and Miguel Ferrer.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 117 : Engineering Ethics and Cyber Law

Batch: 25th | Semester: Fall 2023

Date: 08/12/2023 | **Midterm Examination** | Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

1.	(a)	Define the following terms: i. Ethics ii. Moral iii. Profession	CLO2	3
	(b)	What are the aims of studying Ethics? Briefly Describe.	CLO2	3
2.	(a)	Briefly explain the sources of Morality.	CLO2	3
	(b)	Write the differences between Ethics and Morality.	CLO2	3
3.	(a)	What is Engineering Ethics?	CLO2	2
	(b)	Mention and describe the different types of inquiries in solving Ethical problems.	CLO1	4
4.	(a)	What are the situations leading to Moral Dillema? Briefly Explain.	CLO3	3
	(b)	Describe the Kohlberg Theory of Moral Development.	CLO1	3
5.	(a)	Briefly describe the types of Utilitarianism.	CLO2	2
	(b)	What are the steps in applying the Cost-benefit test in Utilitarian Ethical Theory?	CLO1	4
6.	(a)	What is the need for Judgment in Engineering Standards?	CLO1	2
	(b)	Briefly Explain the types of Engineering Standards that is required to be satisfied by the work of an Engineer.	CLO2	4
7.	(a)	What is the relationship between Responsibility for Harm and Causation of Harm?	CLO1	2
	(b)	Describe the Legal Liability and Moral Responsibility in different scenarios of causing harm.	CLO3	4

HAMDARD UNIVERSITY BANGLADESH

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Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Semester: Fall 2023

Batch: 25^{th} | Time: 2 hours

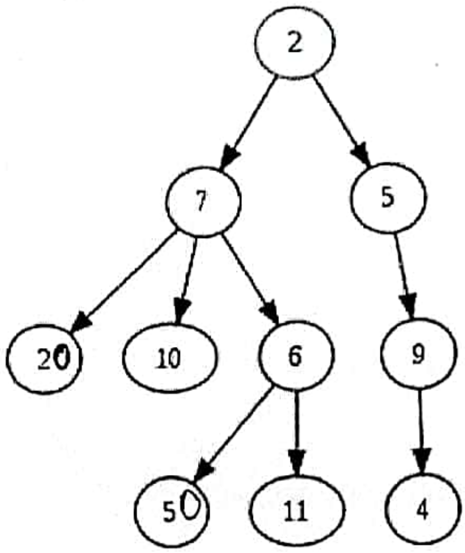
Date: 20/03/2024 | **Final Examination** | Total Marks: 40

$5 \times 8 = 40$

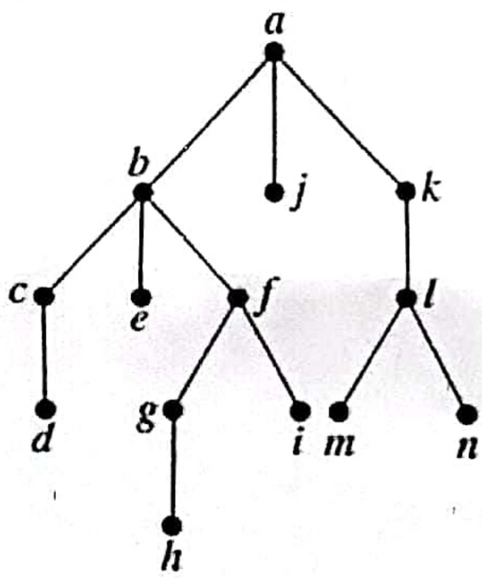
Answer any **five** questions from the following.

1	(a)	State the types of quantification.	CLO1	2
	(b)	Let L(x) denote the statement "Book x is overdue for return at the library." Suppose that of the books checked out, only "Quantum Mechanics" and "Ancient History" are overdue. What are the truth values of L("Quantum Mechanics"), L("Modern Physics"), and L("Ancient History")?	CLO1	3
	(c)	What is the truth value of $\exists x P(x)$, where P(x) is the statement "$x^2 > 10$" and the universe of discourse consists of the positive integers not exceeding 4?	CLO1	3
2	(a)	Define binding variable.	CLO1	2
	(b)	Let P(x) denote the statement "x > 3." What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?	CLO1	3
	(c)	Let Q(x, y, z) be the statement "x^y = z". What are the truth values of the statements $\forall x \forall y \exists z Q(x, y, z)$ and $\exists z \forall x \forall y Q(x, y, z)$, where the domain of all variables consists of all positive real numbers?	CLO1	3
3	(a)	Identify the methods for representing a set.	CLO4	2
	(b)	What is the Cartesian product of X = {x, y} and Y = {3, 4, 5}?	CLO4	3
	(c)	Sketch a Venn diagram that depicts P, the set of prime numbers less than 10.	CLO4	3
4	(a)	Define one-to-one function.	CLO4	2
	(b)	Consider h_1 and h_2 as functions from R to R where $h_1(x) = x^3$ and $h_2(x) = 1 - x$. What are the functions $h_1 + h_2$ and h_1h_2?	CLO4	3
	(c)	Consider g as the function from {w, x, y, z} to {5, 6, 7, 8} where g(w) = 6, g(x) = 5, g(y) = 8, and g(z) = 7. Is g a bijection?	CLO4	3
5	(a)	Describe the division algorithm.	CLO3	2
	(b)	Apply the definition of addition and multiplication in Z_m to find $5 +_7 6$ in Z_7 and $5 *_7 6$.	CLO3	3
	(c)	Calculate the greatest common divisor of 102 and 270 using the Euclidean algorithm.	CLO3	3
6	(a)	Define Abelian group.	CLO4	2
	(b)	Proof the axiom "Distributivity of multiplication over addition".	CLO4	3
	(c)	Justify if (Z, +) a group?	CLO4	3

(a)	Define forest.	CLO4	2
(b)	Find the parent, children, ancestors, siblings and descendants of node 7 in the following tree:	CLO4	3



(c)	Perform pre-order, in-order and post-order traversal of the following tree and list the sequence of nodes visited:	CLO4	3



Hamdard University Bangladesh

Dept. of Electrical and Electronic Engineering (EEE)

Final Examination (Fall'23)

EEE 117 Electrical Engineering

Time: 2 hours Marks: 40

Answer to the question no. 1 and 2 (must) and any 2(two) of the questions among question no. 3, 4 and 5 from the following sets of questions

1. (a) Define the following terms. (3) CLO 1

   i. Alternating current
   ii. Average value
   iii. Rms value

   (b) If $v_1 = -10\sin(\omega t - 30°)$ and $v_2 = 20\cos(\omega t + 45°)$. Find the sum of the two sinusoids. Express the result in time domain. (3) CLO 1

   (c) Calculate the phase angle between $v_1 = -10\cos(\omega t - 50°)$ and $v_2 = 12\sin(\omega t - 10°)$. Transform the sinusoids v_1 and v_2 into phasor. (4) CLO 1

2. (a) Use superposition theorem to find v_x in the circuit of Fig. 2(b). (4) CLO 3

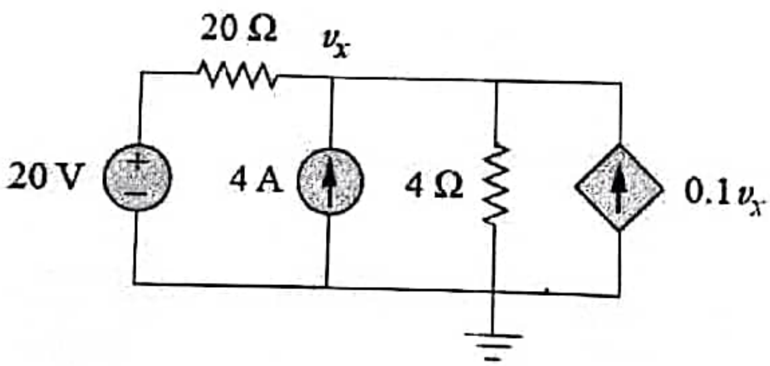


Fig. 2(b)

   (b) Write the statement of the following theorems, (6) CLO 1

   i. Thevenin's theorem — 1
   ii. Maximum power transfer theorem — 1
   iii. Superposition Theorem

3. (a) Use source transformation to find i_x in the circuit shown in Fig. 3(a). (5) CLO 3

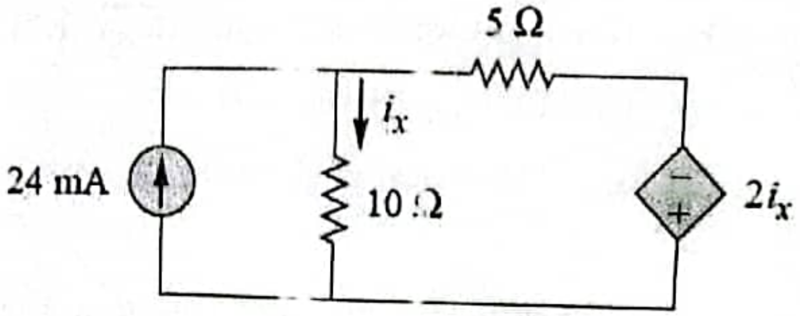


Fig. 3(a)

(b) Find the Thevenin equivalent circuit of the circuit shown in Fig. 3(b). Find the current I using the equivalent circuit. (5) CLO 3

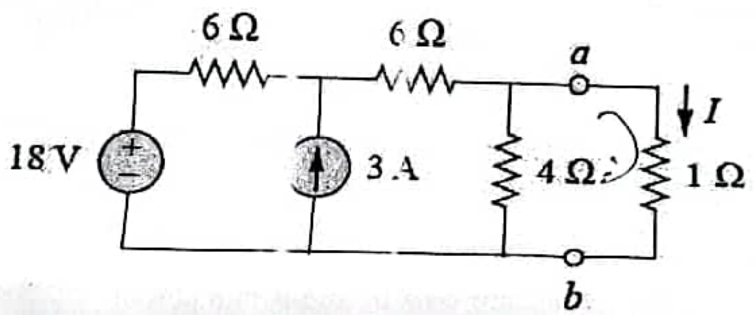


Fig. 3(b)

4. (a) Using mesh analysis, find I_o in the circuit of 4(a). (5) CLO 3

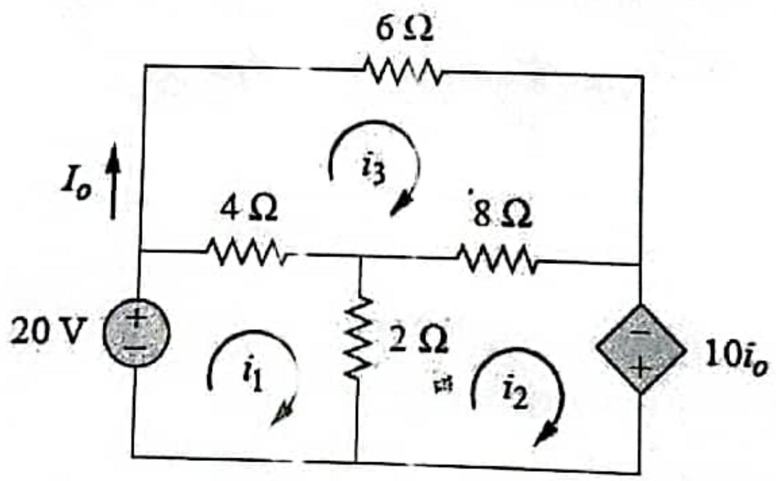
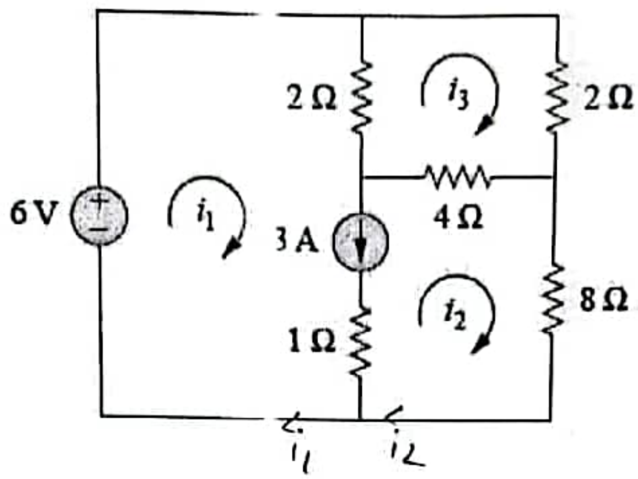


Fig. 4(a)

(b) Use mesh analysis to determine i_1, i_2 and i_3 in the circuit of Fig. 4(b). (5) CLO 3



$i_1 = 3.58$

i_2 $\frac{11}{19}$

$i_3 =$

Fig. 4(b)

5. (a) Find the Norton equivalent circuit of the circuit in Fig. 5(a) at terminals a-b. (5) CLO 3

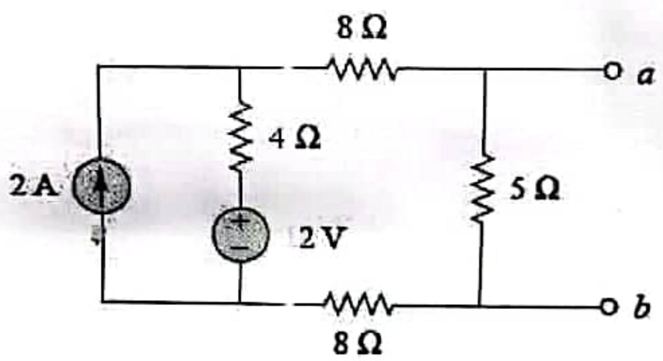


Fig. 5(a)

(b) Find the value of R_L for maximum power transfer in the circuit of Fig. 5(b). Find the maximum power. (5) CLO 3

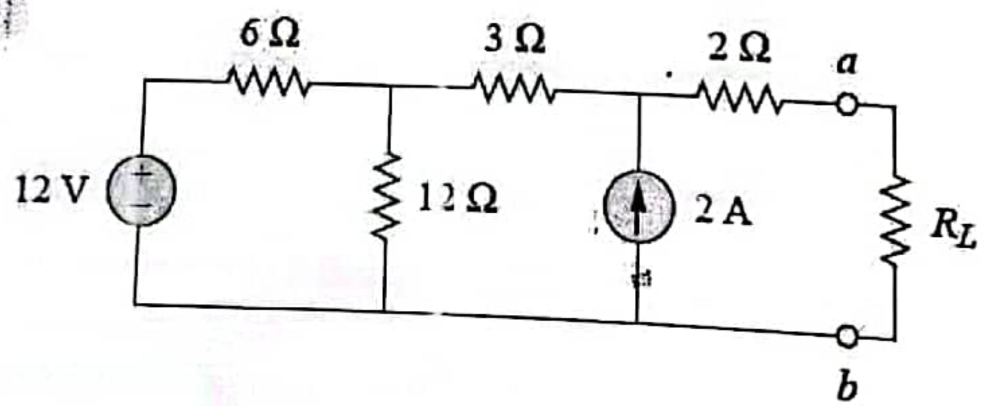


Fig. 5(b)

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Course Code: ENG111
Course Title: English Basics/ English for Technical Communication
Semester: Fall 2023
Final Examination

Batch: 25th
Date: 13 March 2024

Semester: 1st
Time: 2 hours
Total Marks: 40

Part-A: Read the following passage and complete the tasks (1—3)*

The realm of Artificial Intelligence (AI), at the crossroads of computer science, mathematics, and cognitive psychology, is reshaping our world in profound ways. From powering recommendation systems that anticipate our preferences to enabling autonomous vehicles that navigate our streets, AI is steadily permeating every facet of our existence. The potential of AI is expansive and multifaceted. In finance, AI algorithms analyze vast datasets with lightning speed, informing investment decisions and managing risks. In healthcare, AI aids in disease diagnosis, drug discovery, and personalized treatment plans, potentially revolutionizing patient care. In education, AI-powered tutoring systems adapt to individual learning styles, enhancing educational outcomes for students worldwide. Yet, as AI continues to advance, it brings forth a plethora of inquiries. How will AI's rise affect employment patterns as it increasingly automates routine tasks? How do we guarantee the ethical and equitable development and deployment of AI technologies? And what ethical dilemmas emerge as AI systems begin to exhibit intelligence and decision-making capabilities akin to, or even surpassing, human cognition? Just as robotics has prompted reflection on these questions, the ascent of AI compels us to grapple with the implications of its integration into our societies and daily lives. As we navigate this landscape, it becomes imperative to shape AI's trajectory with foresight, responsibility, and a keen awareness of its potential impacts on individuals and societies alike.

No.	Question/ Statement	CLOs	Marks
1.	**Imagine you are two researchers discussing the future of AI. Create a dialogue where you discuss:** A. The potential benefits and challenges of AI in various fields. B. Your personal opinions on the ethical considerations surrounding advancement of AI. C. Your predictions for the future of AI in education and research.	CLO5	1x10=10
2.	**In the following sentences, identify and correct any subject-verb agreement errors:** A. The AI system were analyzing the data from the sensors in the autonomous vehicle. B. One of the researchers believe that AI-driven diagnostic tools is enhancing medical diagnoses.	CLO1	1x5=5

C. The group of scientists have developed a groundbreaking AI algorithm for predicting weather patterns.
D. AI algorithm are constantly learning and evolving, adapting to new data and scenarios.
E. The future of AI seem promising, with the potential to address numerous societal and environmental challenges.

3. **Rewrite the following sentences using the instructions provided in parentheses:** — CLO1 & CLO2 — 1x5=5

A. AI has become a crucial tool in education and research. (Interrogative)
B. ChatGPT is best AI tool in current time. (Negative)
C. We should not use AI in unethical work. (Imperative)
D. Artificial intelligence is the beginning of new era. (Negative)
E. AI will never replace human work. (Interrogative)

Aseen

Part-B: Answer the following questions

4. Imagine you want to be a lecturer in your department of a reputed university. **Write a compelling cover letter expressing your enthusiasm, relevant skills, experience and why you are the ideal candidate for the particular lecturer position.** Highlight how your **academic background** and **extracurricular experiences** have prepared you for this role. Additionally, **outline your objectives** and what you hope to contribute to the team. — CLO4 — 1x8=8

5. Suppose you are writing a CV. You have already written your skills, competencies and Co-curricular and extra-curricular activities. Now, write down- — CLO4 — 3x4=12

i) Career objective (aim, focus, etc.)
ii) Education
iii) References (at least two)

HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering -lan
Faculty of Science, Engineering and Technology

Course Code: PHY111 Course Title: Physics I

Final Examination
Date: 3/11/2024 Total Time: 2 HR Semester: Fall 2023
Total Marks: 40

Answer any four (04) out of the following questions: $4 \times 10 = 40$
[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a) Show that the total energy of a particle executing simple harmonic motion can be represented by the expression $2\pi^2 ma^2 n^2$. **CLO1** **[6]**
   (b) The displacement of a particle executing simple harmonic motion is given by equation y=0.3sin20π(t+0.05) where time t is in second and displacement y is in meter. Calculate the values of amplitude, time period, initial phase and initial displacement of the particle. **CLO1** **[4]**

2. Derive Smith-Helmholtz Equation in optics. **CLO2** **[10]**

3. (a) What is meant by optical aberration? Discuss spherical aberration and astigmatism. **CLO2** **[6]**
   (b) Light travels from air into an optical fiber with an index of refraction of 1.44. (a) In which direction does the light bend? (b) If the angle of incidence on the end of the fiber is 22°, what is the angle of refraction inside the fiber? **CLO2** **[4**

4. (a) What are the conditions of interference of light? **CLO2** **[4]**
   (b) Show that the distance between any two consecutive bright or dark fringes in Young's double slit experiment can be expressed by $\frac{\lambda L}{d}$. **CLO2** **[6]**

5. (a) From Brewster's law show that the reflected and the refracted rays are right angles (90^0) to each other. **CLO2** **[7]**
   (b) A beam of light strikes the surface of a plate of glass with a refractive index of √3 at the polarizing angle. What will be the ray's angle of refraction? **CLO2** **[3]**

6. (a) Distinguish between heat and temperature. **CLO4** **[4]**
   (b) Clarify the first and second laws of thermodynamics with some examples. **CLO4** **[6]**

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Q4.	**(a)**	State Rolle's Theorem.	CLO1	**[2]**
	(b)	State and prove Lagrange's Mean Value Theorem.	CLO3	**[5]**
	(c)	Examine if Mean Value Theorem applied to $f(x) = x^3 - 3x - 2$ in the interval [-2, 3].	CLO2	**[3]**
Q5.	**(a)**	Define maximum and minimum of a function.	CLO1	**[2]**
	(b)	State and prove Leibnitz's theorem.	CLO3	**[4]**
	(c)	If $y = (\sin^{-1} x)^2$ the prove that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$	CLO2	**[4]**